SEVA FIRST INNOVATION CHALLENGE 2026

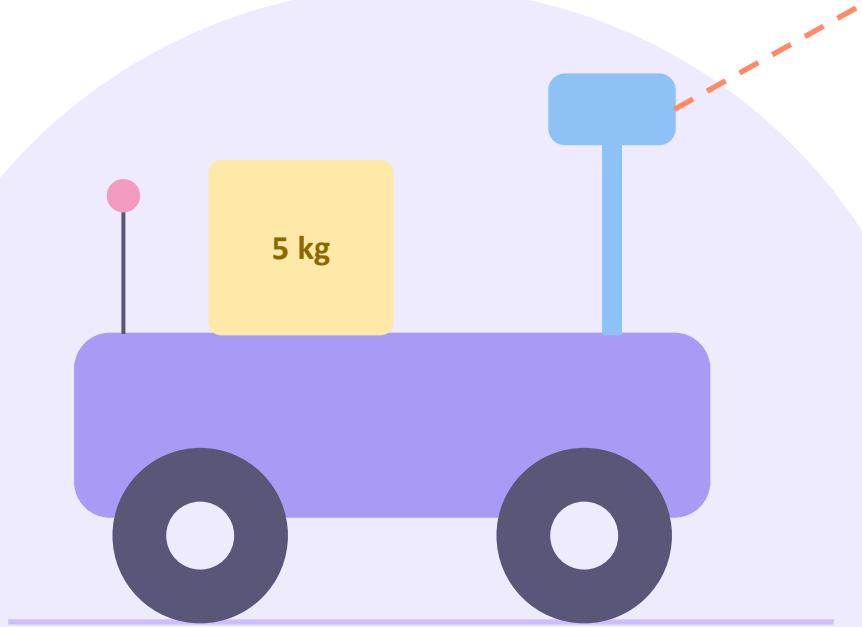
सत्यमेव जयते
Government of the National Capital Territory of Delhi

STAGE 1 · IDEATION · CATEGORY II — DIPLOMA & HIGHER EDUCATION

SAARTHI

Drone-grade, low-latency autonomy for a multi-terrain unmanned ground vehicle

***S**elf-reliant **A**utonomous **A**ll-terrain **R**over with **T**arget & **H**azard **I**ntelligence*



PROBLEM STATEMENT

NAT-003 · R.A.M.A.N

CHALLENGE OWNER

Delhi Technological University

TEAM

SAMAYPUR SHOOTERS

Team details

Team at a glance

TEAM NAME
SAMAYPUR SHOOTERS

CATEGORY
II · Diploma & Higher Education

PROBLEM STATEMENT
NAT-003 · R.A.M.A.N

CHALLENGE AREA
Defence, Intelligence, Space & National Security

INSTITUTION
Delhi Technological University

FACULTY MENTOR
Dr. Anamika Chauhan

TEAM SIZE
3 members

Mohammed Adeeb Khan
B.Tech EE · 3rd year

Team Lead · Electronics, Power & Safety

Electronics & power · E-stop safety chain · Radio links & RTK · Integration

Bhavya Jain
B.Tech EE · 3rd year

Perception & Machine Learning

Datasets · Lane, sign & light models · Face matching · TensorRT on Jetson

Ashmit Bindal
B.Tech IT · 3rd year

Software · Autonomy, Simulation & Testing

ROS 2 & ArduPilot · Nav2 planning · Gazebo twin · Test logs & docs

Detailed problem statement

IN PLAIN WORDS

Security and emergency teams still send people into hazardous zones because no single, affordable ground robot can follow a route, obey traffic instructions, dodge hazards, climb rough terrain and identify a target — all on its own.

OBSERVATION CONTEXT

- **Where:** DTU Sports Complex — outdoor, daylight, multi-stage track
- **Who is exposed:** recon, security and emergency-response personnel
- **Today:** separate or human-assisted systems for each task (PS §7.3)

One platform must do all of this — with no human in the loop

Lane discipline
Zero track-boundary departures

Signs & lights
Identify, interpret and obey

Potholes & obstacles
Detect and avoid, no collision

20° ramp & debris
With the 5 kg payload secured

Face match
Pick the correct printed panel

Laser indication
Eye-safe, ≥ 2.0 s continuous

Hard constraints: no uplink in the autonomous round · 2–10 km/h · 60–200 × 40–120 cm · wireless E-stop ≥ 150 m · licence-exempt bands only

Evidence/data/survey/interviews

What the problem owner reports

- **Gap 1:** no unified hardware + software architecture for multi-environment operation
- **Gap 2:** weak sign and traffic-light compliance while holding lanes
- **Gap 3:** face matching, safe optical indication and hardware safety are rarely integrated
- **Baseline:** separate or human-assisted systems per task

Rule → our design target

Rule (PS)	Our target
Avg. speed 2–10 km/h	4–6 km/h cruise
E-stop range ≥ 150 m	≥ 300 m (2×)
Laser hold ≥ 2.0 s	3.0 s (1.5×)
20° ramp	Tested to 25°
5 kg payload	Tested with 8 kg
0 boundary departures	0 in last 10 mock runs

Primary research (in progress)

- **Organiser clarifications:** arena dimensions, sign catalogue, panel size
- **Campus security staff:** patrol pain points and hazards
- **Civil-defence volunteers:** what a recon robot must report
- **Robotics faculty:** feasibility review of our 75-day plan

Logged with dates and outcomes on slide 13

Source: SEVA FIRST Problem Statement NAT-003 R.A.M.A.N, §5–§12 (Delhi Technological University).

Existing solutions and limitations

Approach	Lane & route	Signs & lights	Ramp & terrain	Face-ID + laser	No human	Low cost
Road-autonomy stacks (Autoware / Apollo-class) Heavy compute, HD maps; assume paved roads	●	●	○	○	●	○
Tele-operated tactical UGVs (e.g. DRDO Daksh-class) Operator in the loop; video latency limits precision	○	○	●	○	○	○
IGVC-style university rovers Lanes + GPS waypoints; no traffic rules or target ID	●	◐	◐	○	●	◐
Open-source GPS rovers (ArduPilot kits) Waypoints only; no vision perception	○	○	◐	○	●	●
Fixed CCTV face recognition Stationary; no mobility or safe indication	○	○	○	◐	●	◐
SAARTHI (ours) All tasks on one stand-alone, fail-safe platform	●	●	●	●	●	●

● full ◐ partial ○ absent

Insight: the pieces exist — the gap is integrating them under hard real-time and safety limits, which drone autopilots already solve.

Proposed technical solution

SAARTHI fuses a proven drone autopilot core with an edge-AI perception brain on a rugged 4WD chassis — one stand-alone platform for every R.A.M.A.N task.

1 · Drone-grade core

- **Pixhawk 6C running ArduPilot Rover:** 400 Hz control loop, EKF3 sensor fusion
- uXRCE-DDS / MAVLink 2 wired link to the AI computer (< 2 ms)
- ExpressLRS 500 Hz control + 5.8 GHz digital FPV for the manual round
- RTK GNSS (u-blox F9P) for cm-level task waypoints

2 · Edge-AI perception

- Jetson Orin Nano Super with TensorRT INT8 models
- Lane and pothole segmentation with a classical cross-check
- Sign and traffic-light detector with temporal voting
- SCRFD + ArcFace face matching driving a laser visual servo

3 · Fail-safe by design

- Hardware E-stop chain that bypasses all software
- **433 MHz LoRa heartbeat E-stop:** link loss means stop
- Autonomous mode physically disables telemetry receive
- Laser fires only on a depth-verified flat printed panel

Platform: 4WD skid-steer · 80 × 60 cm · ~25 kg with payload · 24 V LiFePO4 · independent suspension · 30 cm payload cradle with locating corners and strap

Conceptual architecture

SENSE

- Wide RGB camera · 60 fps
- Depth camera (stereo)
- 2D LiDAR · 360°
- RTK GNSS · u-blox F9P
- IMU + wheel encoders

THINK · Jetson Orin · ROS 2

- Perception models (TensorRT INT8)
- World model + fused costmap
- Mission behaviour tree (task FSM)
- Planner: Nav2 + pure pursuit / MPC
- uXRCE-DDS / MAVLink 2 bridge

ACT · Pixhawk 6C

- EKF3 fusion · 400 Hz loop
- Skid-steer mixer + PID
- Motor drivers → 4WD
- Laser pan-tilt gimbal
- Pitch-aware torque on ramp

RADIO PLAN

- Manual: ELRS 2.4 GHz
- Manual: 5.8 GHz FPV
- Auto: telemetry TX-only
- Auto: RTK RX-only
- E-stop: 433 MHz LoRa

SAFETY CHAIN
hardware only

Mushroom E-stop on vehicle + LoRa E-stop RX (20 Hz heartbeat) → Latching safety relay → Main contactor opens + driver brake → **Stop < 0.5 s, no software in path**

Working principle

1 **Self-check**
Sensors, heartbeat, RTK fix

→ 2 **Lane follow**
Segment → centre-line → pure pursuit

→ 3 **Signs & lights**
Detect, classify, stop / go

→ 4 **Avoid hazards**
LiDAR + vision costmap, replan

→ 5 **Ramp & debris**
Pitch-aware torque feed-forward

→ 6 **Face-ID zone**
GPS trigger, ArcFace match

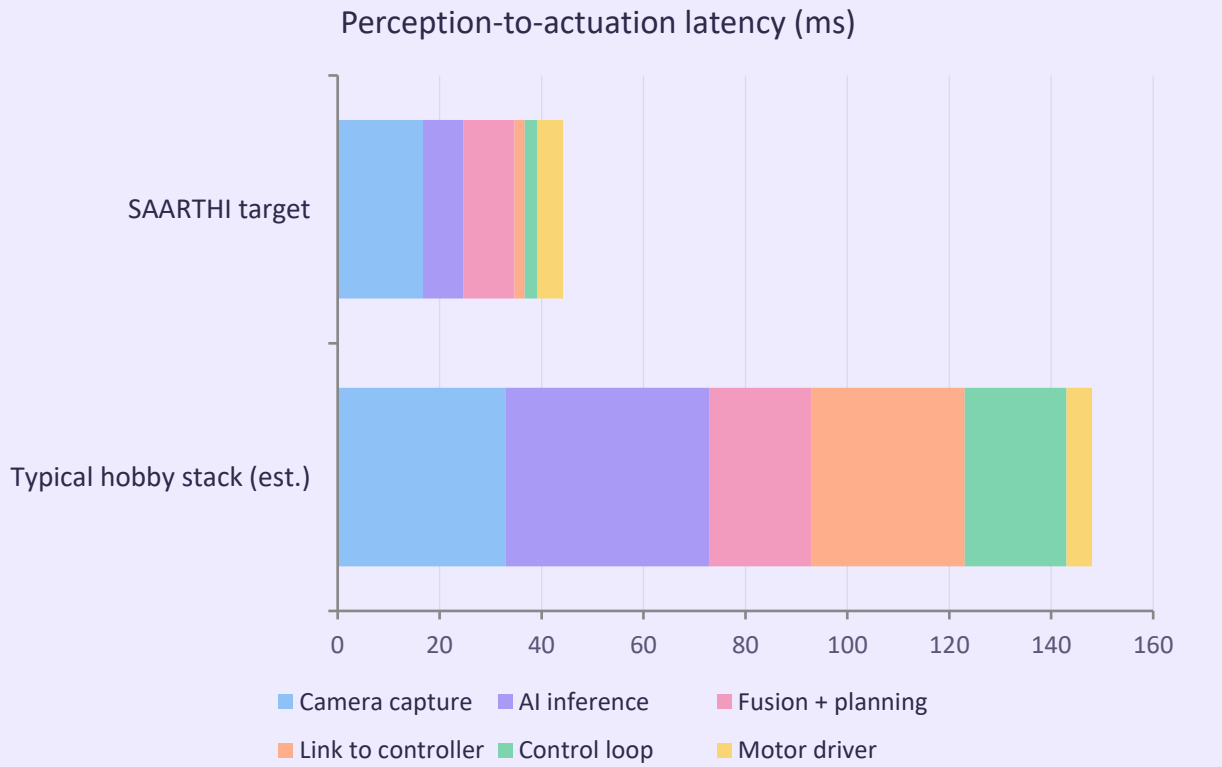
→ 7 **Laser & finish**
Visual-servo hold ≥ 3 s, resume

Edge-ML model stack

Task	Model (edge-optimised)	Training data	Latency*	Safety fallback
Lane + pothole segmentation	TwinLiteNet-style / YOLOv8n-seg	Arena replica + synthetic sim renders	~5 ms	White-mask + Hough-circle check
Road signs & traffic lights	YOLOv8n detector + colour-state voting	IDD / GTSRB pre-train → rulebook signs	~4 ms	3-of-5 frame vote; default = stop
Static obstacles	LiDAR + depth occupancy grid	Geometry, no training needed	~3 ms	LiDAR-only emergency stop zone
Face detect + match	SCRFD + ArcFace (MobileFaceNet)	Public pre-trained; reference enrolled live	~3 ms	Top-1 vs top-2 margin, else re-approach

*Estimated on Jetson Orin, TensorRT INT8, per frame; to be benchmarked in week 5.

Preliminary calculations/simulation



≈ 148 ms → ≈ 44 ms. Simulation: Gazebo digital twin of the arena CAD, 500 randomised runs planned.

Ramp traction

F = mg(sin20° + 0.05·cos20°), m = 25 kg

95 N → ≥ 4.8 N·m per wheel (2× SF)

Tip-over margin

atan(½ wheelbase / CG height) = atan(0.30 / 0.25)

50° limit vs 20° ramp

Stopping distance

$v \cdot t_{lat} + v^2/2a$ at 10 km/h, $a = 3\ m/s^2$

1.4 m vs 4 m sensing range

E-stop link budget

433 MHz LoRa, 10 dBm, FSPL at 300 m = 75 dB

≈ 60 dB fade margin

Endurance

24 V × 20 Ah LiFePO4 = 512 Wh at ~220 W

≈ 2.3 h vs < 20 min run

Laser aim

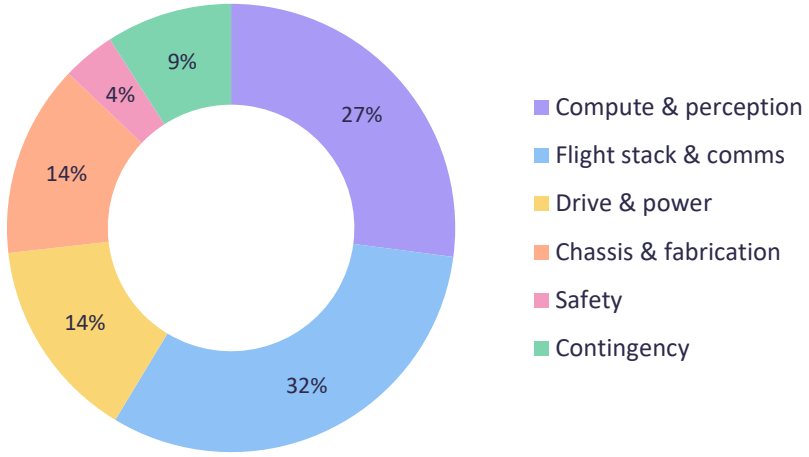
A4 panel at 3 m → ±2.0° window; servo 0.1°

20× aim margin, 3 s hold

Estimated cost

Item	Category	Cost (INR)
Jetson Orin Nano Super (8 GB) dev kit	Compute	₹ 32,000
Depth camera + wide RGB camera	Perception	₹ 38,000
2D LiDAR (360°)	Perception	₹ 8,000
Pixhawk 6C autopilot + GPS mast	Flight stack	₹ 20,000
RTK GNSS pair (2 × u-blox F9P + antennas)	Comms	₹ 45,000
ELRS link + digital FPV + telemetry radio	Comms	₹ 26,000
4 × 24 V geared motors + 2 dual drivers	Drive	₹ 28,000
24 V 20 Ah LiFePO4 + BMS + charger	Power	₹ 14,000
Al-6061 chassis, suspension, wheels	Chassis	₹ 18,000
Wiring, fuses, IP54 enclosures, 3D prints	Fabrication	₹ 22,000
E-stop pair, contactor, Class-2 laser gimbal	Safety	₹ 11,000
Contingency (10 %)	Buffer	₹ 26,200
Total (prototype)		**₹ 2,88,200**

Share of budget



≈ ₹2.9 L

total prototype cost

₹ 0

software licence fees (open stack)

Expected beneficiaries

0

personnel needed inside the hazard zone during an autonomous recon pass

< 50 ms

perception-to-actuation target, ~3× faster than typical hobby stacks (est.)

≈ ₹2.9 L

prototype cost vs ₹20 L+ for imported research UGV platforms (indicative)

1 → N

modular design scales from one prototype to agency fleets

Primary users

Defence, intelligence and national-security agencies — tactical and reconnaissance teams

- Route recon without exposing soldiers
- Autonomy kit that can be retrofitted to larger platforms

Disaster response

NDRF / SDRF and fire services scouting debris zones

Police & municipal security

Campus and perimeter patrols that respect road rules

Infrastructure operators

Solar-farm, rail-yard and pipeline inspection

Robotics researchers & students

Open, documented stack for Indian teams

Initial development plan

G0 D1–15

Digital twin

Arena replica in Gazebo from CAD; v1 models on synthetic data

EXIT CRITERION
Lane-follow in sim, 20 clean runs

G1 D10–30

Subsystem bench

E-stop chain, drive, power and sensors tested on the bench

EXIT CRITERION
Stop < 0.5 s at 300 m E-stop range

G2 D25–40

Manual qualification

ELRS + FPV + GCS; ramp and payload trials

EXIT CRITERION
5 clean manual laps on FPV only

G3 D36–60

Autonomy integration

Lanes, signs, hazards and face-ID running on the vehicle

EXIT CRITERION
Each task ≥ 90 % success in mock arena

G4 D56–75

Competition ready

Full-course mock runs; inspection checklist

EXIT CRITERION
30 full runs, 0 departures in last 10

Principles: simulate first · build the E-stop before the motors spin · every feature gets a pass/fail metric · weekly full-integration run with logged telemetry

Evidence of user/problem interaction

#	Stakeholder	Mode	What we validate	Status
1	DTU UGV technical panel	Clarification query	Arena dimensions, sign catalogue, face-panel size	Planned
2	Campus security supervisors	On-site interview	Patrol routes, hazards, what a robot must report	Planned
3	Civil-defence / NDRF volunteer	Phone interview	Recon needs in debris and restricted zones	Planned
4	Robotics faculty mentor	Design review	Feasibility of the 75-day plan and budget	Planned
5	Mock-arena field trial	Field test	Lane and sign detection in Delhi daylight glare	Planned

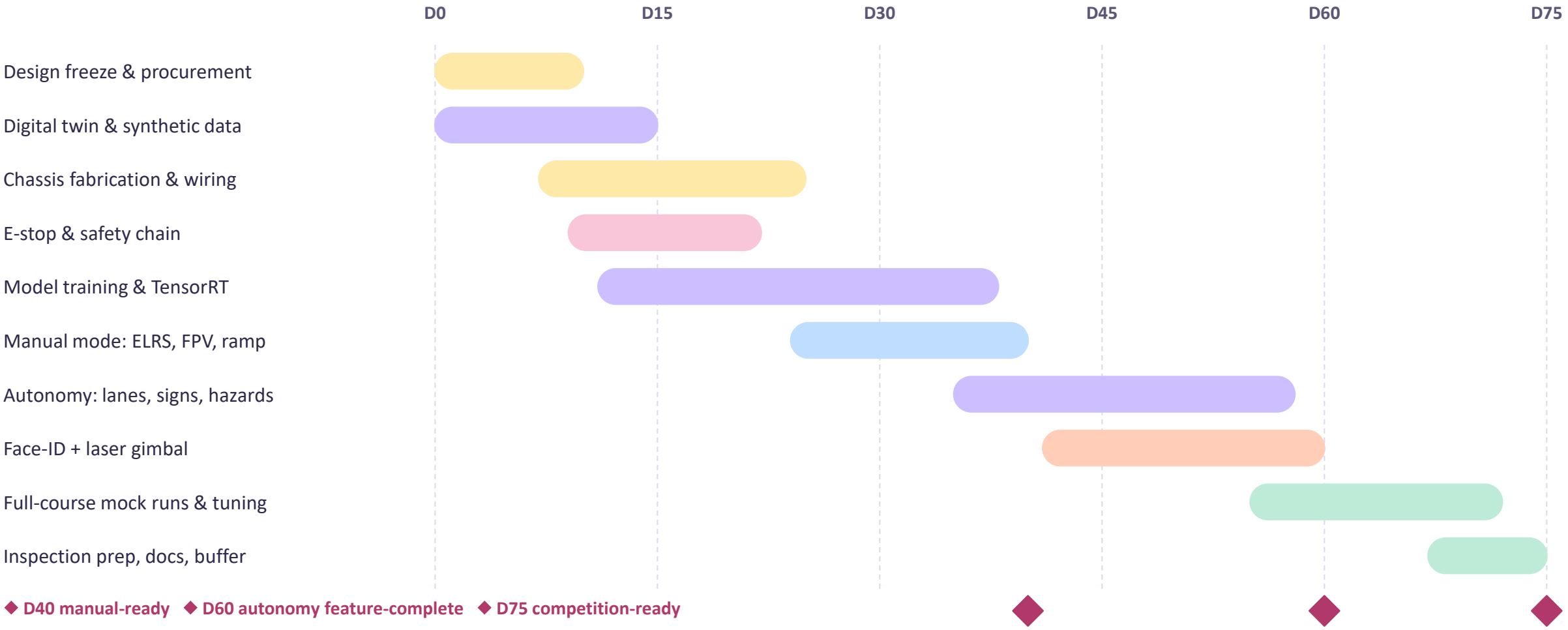
Already done: close reading of the NAT-003 problem statement, turned into the measurable targets on slide 4.

How we record it

- Dated interview notes with name and role (with consent)
- Field photos and short videos of trials
- Run logs with telemetry for every test
- Survey form for security and response staff

Attached as optional supplemental documents (Points 3 and 5).

75-day work plan



Team composition and roles

Mohammed Adeeb Khan
B.Tech EE · 3rd year

Team Lead · Electronics, Power & Safety

- System architecture and integration
- Battery, BMS, motor drivers, harness
- Pixhawk wiring, sensors, laser driver
- ELRS, FPV, RTK and telemetry links
- Wireless + mechanical E-stop chain
- Chassis build and payload cradle

Bhavya Jain
B.Tech EE · 3rd year

Perception & Machine Learning

- Datasets, labelling, synthetic data
- Lane and pothole segmentation
- Sign and traffic-light detection
- SCRFD + ArcFace face matching
- TensorRT INT8 deployment on Jetson
- Laser visual servo and aim loop

Ashmit Bindal
B.Tech IT · 3rd year

Software · Autonomy, Simulation & Testing

- ROS 2 stack and MAVLink / DDS bridge
- ArduPilot Rover tuning, EKF3
- Nav2, behaviour tree, mission logic
- Gazebo digital twin and replay tests
- Test matrix, run logs, telemetry GCS
- Compliance and PDR documentation

Faculty mentor: Dr. Anamika Chauhan · Delhi Technological University

Resource/facility requirements

Hardware (on vehicle)

- Jetson Orin Nano Super
- Pixhawk 6C autopilot
- 2 × u-blox F9P RTK GNSS
- Depth + RGB camera, 2D LiDAR
- ELRS + digital FPV kit
- 433 MHz LoRa E-stop pair

Software (all open source)

- ROS 2 Humble + Nav2
- ArduPilot Rover, MAVLink 2
- Gazebo / Isaac Sim
- PyTorch → TensorRT
- InsightFace (ArcFace, SCRFD)
- Mission Planner / QGC as GCS

Facilities

- Mechanical workshop
- 3D printing and laser cutting
- Electronics bench and soldering
- Open ground ≥ 300 m for E-stop tests
- GPU workstation for training

Requested from organisers

- Arena CAD and track dimensions
- Road-sign and traffic-light catalogue
- Face-panel size and gallery format
- Base-station power and GCS specification

Entry and Target TRL, if any



Entry · TRL 3

Proof of concept: each critical function — lane segmentation, ArcFace matching, ArduPilot Rover control, LoRa E-stop — is proven on its own in literature and open-source projects.

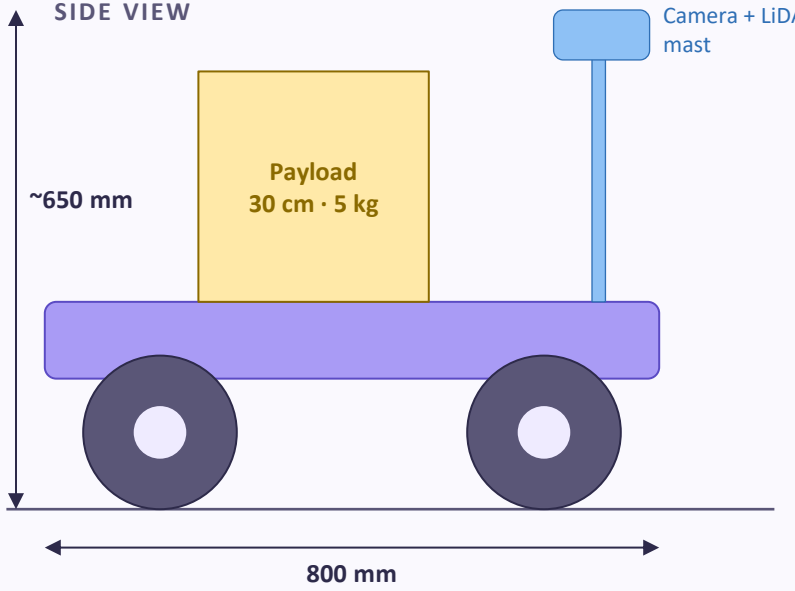
Climb · TRL 4 → 5

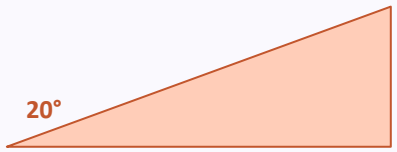
TRL 4 by D30: subsystems integrated on the bench. TRL 5 by D60: full vehicle validated in our mock arena.

Target · TRL 6

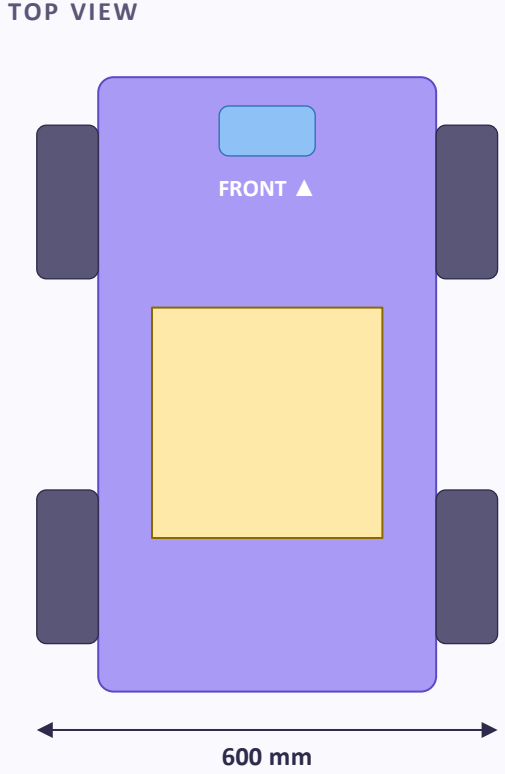
System prototype demonstrated in a relevant environment: the complete UGV running the DTU multi-environment track autonomously.

Technical drawings/calculations/simulation





Ramp: 95 N tractive force, ≥ 4.8 N·m per wheel (2× SF)



Schematic, not to final scale. Full CAD in the supplemental archive.

Parameter	Design value
Overall L × W × H	800 × 600 × ~650 mm
Mass with payload	~25 kg
Drive	4WD skid-steer, 4 × 24 V geared motors
Wheels	200 mm, independent swing-arm suspension
Speed	4–6 km/h cruise, firmware cap 9.5 km/h
Battery	24 V 20 Ah LiFePO4 (512 Wh), BMS + fuse
Compute	Jetson Orin Nano Super + Pixhawk 6C
Ground clearance	≥ 80 mm for debris
Payload cradle	Printed corner locators + ratchet strap

Literature/patent survey

Work	Key idea	How SAARTHI uses it
Redmon et al., CVPR 2016; Ultralytics YOLOv8, 2023	Single-stage real-time object detection	Sign and traffic-light detector
Che et al., 2023 — TwinLiteNet	Tiny drivable-area and lane segmentation	Lane and pothole mask on Jetson
Deng et al., CVPR 2019 — ArcFace	Angular-margin face embeddings	1-to-N printed-panel matching
Guo et al., 2021 — SCRFD	Efficient face detection	Locating faces on panels
Macenski et al., IROS 2020 — Nav2	Behaviour-tree navigation, costmaps	Planner and mission tree
Varma et al., WACV 2019 — IDD	Indian unstructured driving dataset	Pre-training for Indian signs
ArduPilot, MAVLink 2, ExpressLRS	Drone autopilot and low-latency RC link	Drone-grade control core
IGVC rules and design reports	UGV lanes, obstacles and GPS benchmark	Test methodology

Patent search plan: InPASS (Indian Patent Office) + Google Patents for combined UGV traffic-compliance, face matching and safe laser designation; focus on our interlock claims (slide 21).

Laboratory/fabrication access

Mechanical workshop

Chassis cutting, drilling, welding

Source: Institute workshop

To confirm

3D printing & laser cutting

Brackets, payload locators, sensor mounts

Source: Institute makerspace / local vendor

To confirm

Electronics lab

Harnessing, soldering, power testing

Source: Institute electronics lab

To confirm

CNC & sheet metal

Aluminium plates and brackets

Source: Fabrication vendors, Delhi NCR

To confirm

Open test ground

E-stop range, speed and ramp trials

Source: Institute sports ground

To confirm

GPU compute

Model training and simulation (never on-vehicle)

Source: Team workstation / cloud GPU

To confirm

Novelty and IPR potential

Drone-grade latency core on a UGV

Autopilot + DDS + ELRS + digital FPV: < 50 ms autonomy loop and < 40 ms manual control-to-wheel.

Compliance-by-design radio

A physical switch disables telemetry receive when autonomy is armed; one-way telemetry is enforced in hardware and checkable at inspection.

Depth-verified laser interlock

Laser enables only on face match AND flat printed surface (depth) AND inside the face-task geofence.

Heartbeat fail-safe E-stop

Independent 433 MHz LoRa link; 200 ms of silence means stop. No software in the path.

IPR strategy

- **Provisional patent:** depth-verified safe optical designation method
- **Design registration:** modular payload cradle and sensor bay
- **Copyright:** SAARTHI integration software (open core)
- **Publication:** benchmark paper after the event

References

1. SEVA FIRST Problem Statement NAT-003 R.A.M.A.N, Delhi Technological University, 2026.
2. DoT / WPC Wing, Govt. of India — licence-exempt frequency band notifications.
3. The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023).
4. IEC 60825-1 — Safety of laser products.
5. ISO 13850 — Safety of machinery: emergency stop function.
6. J. Redmon et al., "You Only Look Once: Unified, Real-Time Object Detection," CVPR 2016.
7. J. Deng et al., "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," CVPR 2019.
8. J. Guo et al., "Sample and Computation Redistribution for Efficient Face Detection" (SCRFD), 2021.
9. Q.-H. Che et al., "TwinLiteNet: An Efficient and Lightweight Model for Driveable Area and Lane Segmentation," 2023.
10. S. Macenski et al., "The Marathon 2: A Navigation System," IROS 2020.
11. S. Macenski et al., "Robot Operating System 2: Design, Architecture, and Uses in the Wild," Science Robotics, 2022.
12. G. Varma et al., "IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments," WACV 2019.
13. ArduPilot Rover documentation; MAVLink 2 protocol specification.
14. ExpressLRS open-source RC link documentation.
15. Intelligent Ground Vehicle Competition (IGVC) official rules.
16. u-blox ZED-F9P integration manual (RTK GNSS).